

DIFFUSION-MODELS AS INPAINTING INQUIRIES FOR EXPRESSIVE-ART BASED FORMATIVE-ASSESSMENT

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Abstract

This work-in-progress paper presents our expressive-art based **Formative-Assessment (FA)** practice as inquiry in intercultural settings for liberation STEM education. The settings focus on target groups of repressed adult-migrants/students, migrated to Germany, due to the enduring oppressive structures and the current wars in Ukraine and Syria. The target groups are interested in experimenting and programming Diffusion-Model (DM) inpainting algorithms to restore their damaged cultural or/and personal images due to the wars.

The paper shows how the practice is conceptualized to enable two DM prototypes.

- 1 A prototype for Ukrainian target group: It is conceptualized to serve a target group of Ukraine students and help them experiment DM inpainting algorithms to face and overcome the repressive barriers in Ukrainian cultural context.
- 2 A prototype for Syrian target group: It is conceptualized to serve a target group of Syrian students and help them experiment DM inpainting algorithms to face and overcome the repressive barriers in Syrian cultural context.

For enabling our expressive-art based FA workshops to approach the cultural aspect of repression in context of each target group, the paper shows how the workshops' two DM prototypes are grounded based on the theoretical concepts "Generative words", "Generative themes", "Critical consciousness" and "Reading the world precedes reading the word" suggested by the critical educator Paul Freire [1,2,3].

The paper also shows the differences between the DM inpainting algorithms used to initialize and build the two DM prototypes and the deep-learning and machine learning algorithms used in our previous research works [4,5,6,7,8].

To test the two DM prototypes, test cases are suggested as next steps of this work-in-progress paper. Finally, for reflection and feedback exchange with the ICERI community we summarize the limits of the two DM prototypes and our next steps to improve them.

Keywords: Formative Assessment, Liberation STEM education, Expressive Art.

1 INTRODUCTION

Assessment is usually understood to have two purposes: summative and formative. Summative assessment is used to judge students' achievements at the end of an instructional unit, whereas formative assessment aims to gather evidence about students' learning progress to influence and guide teacher's approaches, pedagogies, and priorities. According to Hattie¹, using two notions (formative vs summative) is not helpful in educational debate about assessment and visible learning [9]. To improve the debate Hattie suggests using the single notion "Assessment-capable learner". This notion enables him to argue that both forms of assessments: summative and formative are important. To support his argument, he used the metaphor suggested by Robert Stake² that draws not only a clear distinction line between formative and summative assessments but also highlights the importance of them both. The metaphor is: "*When the cook tastes the soup, that's formative and when the guests taste the soup, that's summative*". Hattie's approach of Visible Learning [9] was developed in context of primary schooling, therefore in our research investigations we aim to test if his pedagogy and its explaining constructs hold or do not hold in contexts of our offers for groups of freshman students or/and migrant students in a first year STEM university program. Our works in [4,5,6,7,8] are about research based **Formative Assessment (FA)** tasks in STEM university contexts. They are about how to understand the pedagogy-assessment linkage and

¹ John Allan Clinton Hattie (born 1950) is a New Zealand education academic. He has been a professor of education and director of the Melbourne Education Research Institute at the University of Melbourne, Australia, since March 2011.

² Robert E. Stake (born December 18, 1927) is a Professor Emeritus of Education at the University of Illinois, Urbana-Champaign. https://en.wikipedia.org/wiki/Robert_E._Stake, retrieved on 10.07.2024.

how to approach the research based FA interventions/tasks contextualized in different STEM teaching or training settings. Examples of these tasks are: a task contextualized in workshops for developing deep-learning APIs [4], a task contextualized in training projects for big-data analytics [5], and a task contextualized in a computer organization course that has about 130 freshman students [6]. In contrast to our works in [4,5,6,7,8] that do not focus on the target groups of **repressed** adult-migrants/students, this paper focuses on two target groups of the repressed; namely target group of Ukrainian students, and target group of Syrian students. That two target groups are students who have been migrated to Germany due to the enduring repressive structures and the current oppressive wars in Ukraine and Syria. To approach and understand the repression in context of the two target groups, we organize interventions in form of expressive-art based FA workshops. The FA workshops with concrete examples are introduced in next subsection.

1.1 Expressive-art based fa workshops

The workshops are designed to help migrant students practise and experience assessment forms **not for grading** but for analysing and assessing the **situations** that hinder their self-expressions for a liberation purpose. In these workshops, beside reporting about what barriers hinder them, the students learn to express solutions for overcoming the barriers. A concrete example is shown in Fig 1.1 and Fig.1.2.

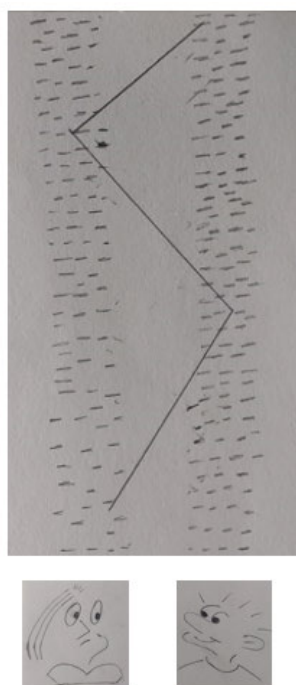


Fig 1.1: Two monologues is not a dialogue.

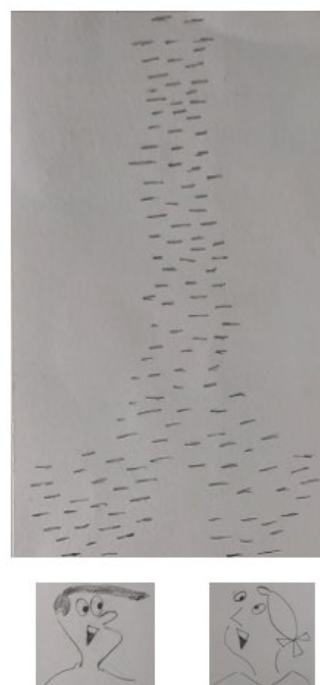


Fig 1.2: Dialogue as mutual process of respect

Figs 1.1 and 1.2 show a concrete conversational situation reported and expressed in drawings by a Syrian migrant student. For this student, the drawing in Fig 1.1 represents one of his lived experiences. The experience outcome is summarized by him as two monologues. And for him “two monologues” is not a dialogue. To express what dialogue means for him, he produces a drawing in Fig 1.2 that shows how a dialogue should be perceived as mutual and respect process that produces shared themes between two equal peers. For this student the experience of dialogue as presented in Fig. 1.2 was not possible in Syrian schooling systems, because the Syrian systems, according to him, are designed to repress the student self-expressions (i.e., the forms of freedom) to maintain the domination and oppression structures that serve the Syrian elites.

The main goal of our FA workshops is to organize and facilitate inquiries in intercultural settings for liberation purpose. For example, the intercultural setting for the student’s lived experience in Figs 1.1 and 1.2 is a setting for Syrian-German interpretations. To put it differently, it is a setting for intersubjective communication. It provides two subjects (i.e., humanists) from two different cultures (e.g., Syrian and German) to express their intersubjectivity as co-humanists interested in inquiry for better humanity. The inquiry is iteration based and its first iteration is initialized by a lived-experience and a situation criticized by a student (e.g., a situation of two monologues criticized by a Syrian student via his drawing in Figs 1.1).

The outcome of the inquiry's initial iterations is used by us as reflections for organizing its upcoming and advanced iterations. The advanced iterations, in contrast to the initial iterations, are driven by interdisciplinary theoretical approaches selected by us during reviewing the literatures in two different disciplines; namely the disciplinary of humanist or critical pedagogy and the disciplinary of modelling **Diffusion-Model (DM)** inpainting algorithms. The approaches selected from the two disciplines constitute our interdisciplinary methodology. The methodology is introduced in next section.

2 OUR METHODOLOGY

Fig. 2 shows our interdisciplinary methodology for inquiry prototype modelling. It is a revised version derived from our methodology in [6]. It is a feedback loop process with two main activities for modelling the inquiry prototypes.

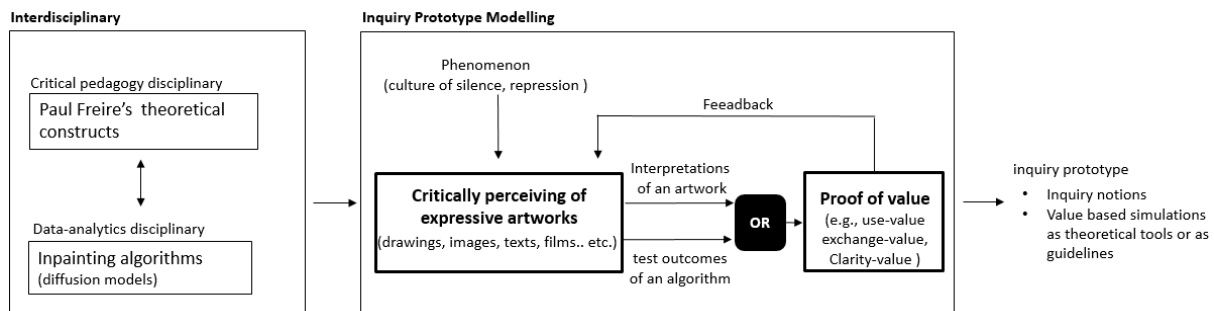


Fig 2. Interdisciplinary methodology for inquiry prototype modelling: revised version derived from [6].

The two activities are:

- Activity for critically perceiving of expressive artworks (i.e., of objects under study): This activity is triggered by a phenomenon of repression reported by a student in the two target groups. A phenomenon of repression is called also a “culture of silence” phenomenon to facilitate formulating and modelling cultural inquiries. Given a student’s expressive drawing (e.g., Fig 1.1), the activity uses this expressive drawing as input to facilitate critical reflection on the student’s perceptions. The reflection is conducted via dialogue between lab instructor(s) and student (s) and its main goal is helping students and lab instructors critically analyze their cultural interpretations in their intercultural workspace setting. The expressive language of the lab instructors/organizers, like the expressive language of the students cannot exist without thought, and neither thought nor language can exist without a *structure to which they refer* [3]. The structural conditions of thought and language define the “expressive art-work” under study (i.e., the object under study). As seen in Fig. 2 this object can be drawings, paintings, texts, or films, etc. that fit the art interests of a student. For organizing effectively, a reflective dialogue with student(s), the organizers (i.e., the lab instructors) should understand the structural conditions in which the thought and language of the students are dialectically framed. Framing dialectically the thought-language of the students as **referable structures** under study is a task initialized by student(s) and further developed by student(s) and lab instructor(s) in their reflective dialogues for enabling perceiving critically the artwork object (i.e., the referable structures) under interpretation. Finally, the outcomes of this critically perceiving activity, as seen in Fig. 2, can be fed to the proof-of-value activity in two formats: namely.
 - Format for collecting interpretations of an artwork. This format is designed to facilitate documenting the interpretations of an artwork. It enables using codifications and decodifications for theorizing the interpretation experiences of the student(s) and the lab instructor(s). The theorizing process is based on the theoretical or explaining constructs selected from the critical pedagogy literature. In scope of this paper, the theoretical constructs are limited to the constructs suggested by the critical educator paulo freire (see fig.2).
 - And format for collecting test outcomes of an algorithm: this format is designed to facilitate documenting the test outcomes of an algorithm. The test outcomes are documented during a co-programming session with student(s) prepared by the lab instructor(s). The session is prepared based on themes of algorithms selected from data-analytics literature. In scope of this paper, the themes of algorithms are limited to the inpainting algorithms based on diffusion models (see Fig.2).

- Activity for proof-of-value: this activity is triggered by the need to analyze rigorously its inputs; namely the interpretations or/and the test outcomes. Its main goal is to produce an inquiry prototype. An inquiry prototype is to facilitate prototyping value-based simulations for experimenting in intercultural settings. It has two experimental components; the first is to provide workable inquiry notions and the second is to provide value based simulations as theoretical tools or as guidelines. If the inquiry prototype is not ready to be produced, the activity produces feedback to refine the activity for critically perceiving of an expressive artwork. The abstract notion “expressive artwork” is a concrete example of a theoretical notion conceptualized as the outcome of a proof-of-value task. This task, in brief, is to fit the values/stakes of two inquiry stakeholders (i.e., students); the first stakeholder values the starting of his/her inquiry from his/her drawings, whereas the second stakeholder values the starting of his/her inquiry as user of a stable inpainting diffusion tool to fix his/her destroyed images. A “destroyed image” and a “drawing” are concrete instances from the abstract notion “expressive artwork.” Moreover, the “expressive artwork” as workable inquiry notion is conceptualized to provide other values (e.g., use-value, exchange-value, or clarity-value) to a student in his/her inquiry journey to understand a repressive phenomenon in his/her cultural context and then to prototype an expressive artwork (e.g., drawings) for experimenting how to overcome it (see section 3 for concrete examples)

Last but not least, the inquiry prototype is conceptualized by our methodology to provide many valuable functions but its main function is to help the FA practitioners (i.e., lab instructors) in understanding better the **structural conditions** of the thought-language artworks (i.e., objects under study) expressed by a student during his/her inquiry journey for critically perceiving of a repressive phenomenon in his/her cultural context. This understanding is required to help the lab instructors prepare suitable lab interventions that fit the intercultural context of the migrant students. To put it differently, to make his/her lab interventions for liberation purpose responsive and sensitive to the cultural barriers, cultural variations, and cultural differences among the migrant students. For two target groups of migrant students in next section we introduce two DM inquiry prototypes.

3 TWO DM INQUIRY PROTOTYPES

The two DM inquiry prototypes are.

- 1 A prototype for Ukrainian target group: It is conceptualized to serve a target group of Ukraine students and help them experiment DM inpainting algorithms to face and overcome the repressive barriers in Ukrainian cultural context.
- 2 A prototype for Syrian target group: It is conceptualized to serve a target group of Syrian students and help them experiment DM inpainting algorithms to face and overcome the repressive barriers in Syrian cultural context.

Based on our methodology described in Fig. 2, the two DM inquiry prototypes are modelled to enable preparing and organizing expressive-art based FA workshops to approach the cultural aspect of repression in context of each target group. The two DM prototypes are grounded based on the theoretical concepts “Generative words”, “Generative themes”, “Critical consciousness” and “Reading the world precedes reading the word” suggested by the critical educator Paul Freire [1,2,3, 10].

To help students understand and use Freire’s concepts for developing their liberation project of actions and reflections, the concepts are clarified to them, during our workshops, via presenting a few images (see Figs. 3, 4, 5, 6, 7,8 and 9) selected from the image based story by Claudius (in German “bildgeschichte von Claudius”) [10]. The story is about a repression phenomenon and it is a concrete example of a liberation project of actions and reflections. It has a sequence of 24 images used as introduction to facilitate translating Freire’s works from 1972 until 1980 to the German speaking countries. The translation outcome was a book with title “Der Lehrer ist Politiker und Künstler: Neue Texte zu befreiender Bildungsarbeit” [10]. To put it in English “The teacher is politician and artist: New texts on the work of liberating education”.

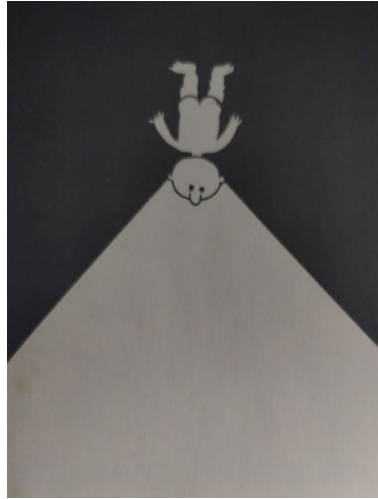


Fig 3. As the world appears today, we need courage to be born [10].

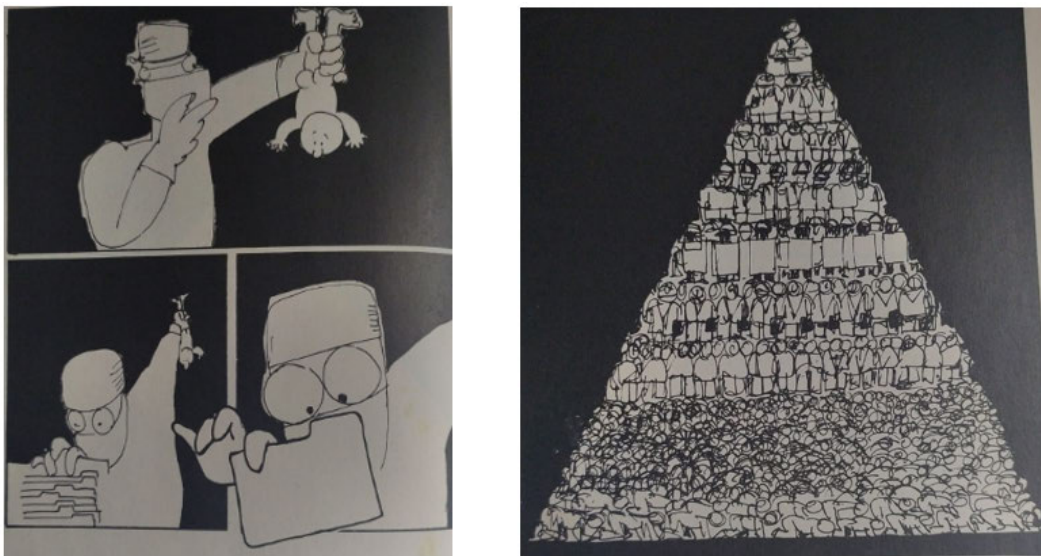


Fig 4. All humans are equal under the law, but some are more equal than others. The social pyramid always has more people at the bottom than at the top [10].



Fig 5. But that is not enough, there are institutions to adapt people to a model of domination (i.e., Herrschaftsmodell) ... and to make them bearers of the system. The two images above are repainted in our workshops based on their origins in [10].



Fig 6. Teacher is a by-product of school institution. He/she reproduces what he/she has been taught [10].

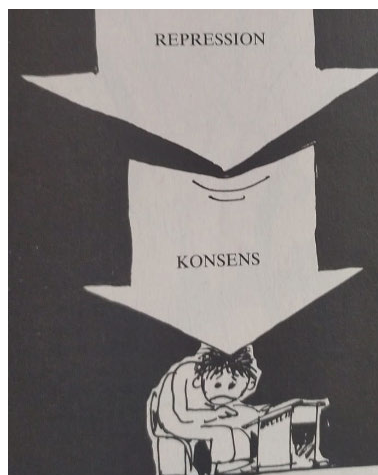


Fig 7. There is a 'political society' made up of the police, the army, the judiciary, etc., which maintains the system through repression. And there is a "civil society" consisting of trade unions, churches, and political parties. Their function is to create social consensus. If both 'political society' and 'civil society' work hand in hand, they can exercise considerable pressure on us. [10].

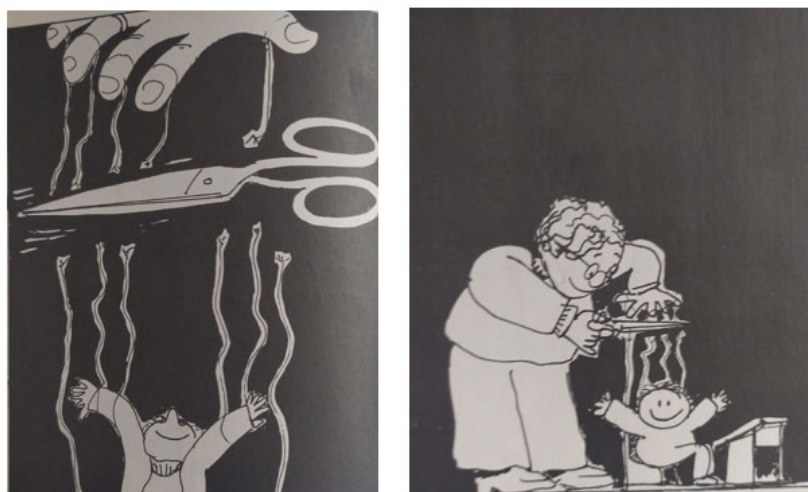


Fig 8. The process of becoming conscious is the consequence of collective praxis. It is the first step towards analysing the reality: questioning our learned knowledge, our preconceived ideas and ideological alienation. A teacher who is himself/herself in the process of liberation can leave his/her students a free space for their own creativity. [10].

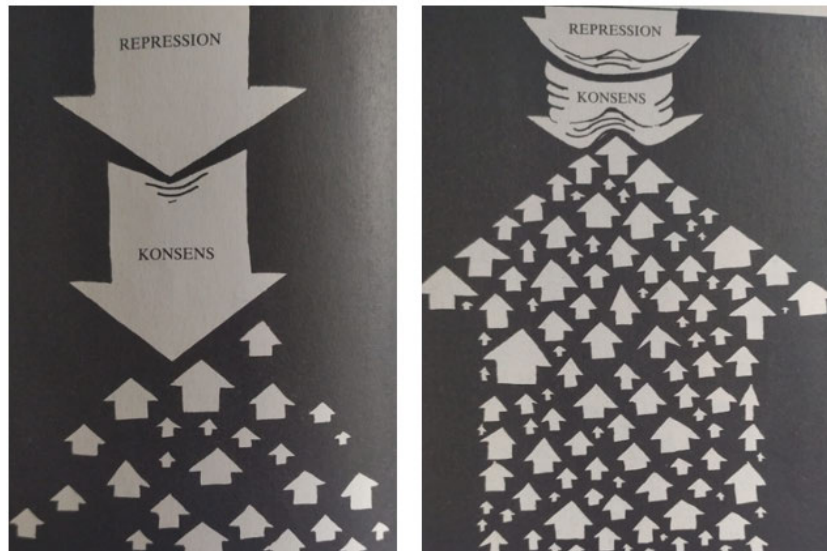


Fig 9. Only in this way, step by step, can new approaches emerge. At first, they are limited to the classroom. But they soon grow beyond the school walls with great vigour. But the repression is still stronger than the new. Small examples, small experiences expand the free spaces. They inspire new experiences in other schools. They are soon taken up by movements outside the school. Influenced by the social movements, they influence them back. The connection between school and life is established. Thus, the free space within the institutions expands [10].

The DM inpainting inquiry prototype operates on images. It takes images as input, and it generates images as output. It is used by the students to generate an image-based story (like story in Figs 1....9) to facilitate their action-reflection-action project. The action-reflection-action project is an iteration-based project to overcome the repressive barriers in their cultural context. As starting point to initialize their project the students have a task to reflect on their lived experiences for preparing or generating images that represent the repressive barriers in their cultural context. Initializing projects for the two target groups of migrant students are discussed in section 4. In next subsection, we present how we use the diffusion models/algorithms as inpainting inquiries for generating images.

3.1 Diffusion models as inpainting inquiries for generating images

A DM inquiry prototype enables a target group of migrant students to look at the diffusion models (i.e., algorithms for inpainting) as inpainting inquiries for generating images. The inquiries should be modelled and conducted to generate images that serve two purposes namely,

- a) To serve representing the repressive barriers under study. The repressive barriers are barriers in the cultural context of a target group. The generated images that represent them serve initializing the image-based story of the target group. To give concrete examples, Figs 6 and 7 in Claudius's image-based story (see section 3) represent repressive barriers.
- b) To serve representing feasible social actions for overcoming a barrier. To give concrete examples, Figs 8 and 9 in Claudius's image-based story (see section 3) represent feasible social actions for overcoming a repressive barrier represented in Figs 6 and 7.

Oriented by the two purposes the students, in their inquiry, should explore the limits of the DM inpainting algorithms and if these kinds of algorithms can help them build their image-based story for their liberation project. In their inquiry the students can play two roles: a role as user of a DM inpainting algorithm and a role as designer of DM inpainting algorithm. In scope of this work-in-progress paper the "user role" is focus whereas the "designer role" will be the focus in our future work. Last but not least, inquiries with the "designer role" will enable us to make comparisons between the DM inpainting algorithms and the deep-learning and machine learning algorithms used in our previous research works [4,5,6,7,8]. These comparisons are planned in our future work.

Finally, for inquiries and experimenting with the “user role”, in next section, we suggest a test case.

4 TEST CASE

The test case, in scope of this work-in-progress paper, is limited to the DM inquiry prototype of the Syrian target group because the group has agreed to share their images (see Figs 10 and 11) publicly for the sake of testing.



Fig 10. Collection of 5 images by a Syrian family. The images show the destruction of their house, street, and life in Damascus due to the current Syrian war started 2011. They are captured in 2014.

The test case is about an inquiry in the intercultural (i.e., Syria-German) setting. During their inquiry the Syrian students aim to express, in image-based story style, their Syrian repressive barriers/issues and how they can contribute to overcome these barriers. Their expressions in images should be prepared and simulated via the help of a DM inpainting tool (e.g., the Stable Diffusion tool available via the online link “<https://getimg.ai/features/inpainting>” or the pencil tool available via the online link “<https://pincel.app/tools/inpaint>”). The students test if using the DM inpainting tool’s functions are helpful or not for restoring their damaged images due to the war. They test also if their image inpainting goal is achievable via the DM inpainting tool’s functions or their goal requires other advanced functions.

For each damaged image in Fig 10 and Fig 11, the students, during their prototyping inquiry, produce inpainting-value based simulations for testing the inpainting steps require to restore the image. The results of the test steps should be reported by the students in a format that enables sharing their reflections on how they perceive their repressive barriers. The question “are their perceptions critical enough for enabling the self-overcoming of the barriers?” drive the further steps of their inquiry. Finally, the ten images in Fig. 11, from a project by the Syrian sculptor Assem Al-basha, have Syrian captions to clarify why the sculptor decided in Granada (Spain) in 2018 to start his act for restoring the damaged Al-Ma’arri sculpture. The captions will be translated to English during our oral presentation. To facilitate the presentation, we present a concrete example of test result in next section.

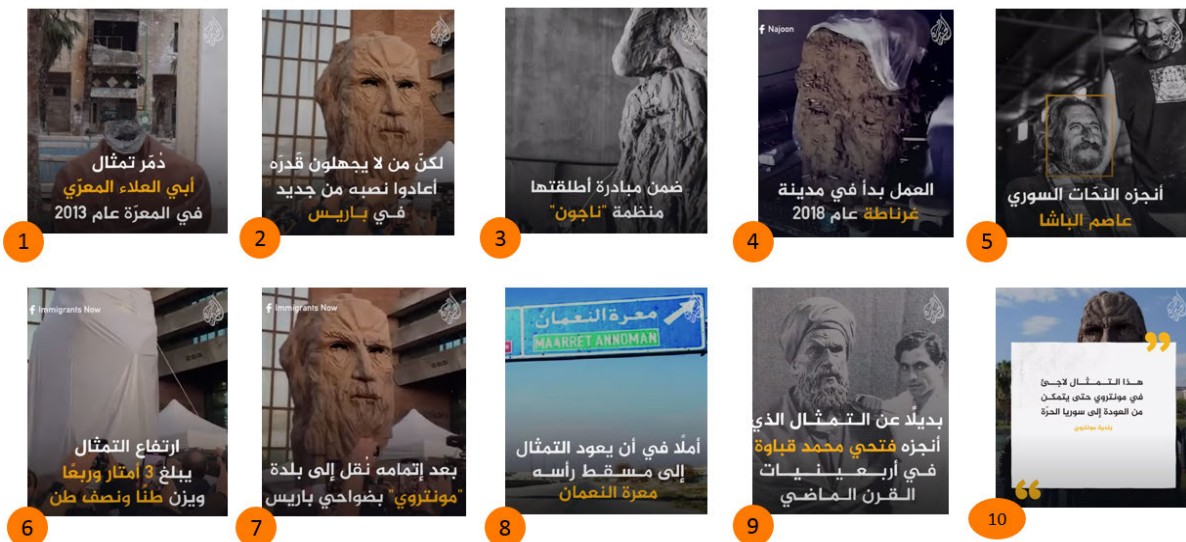


Fig 11. Collection of 10 images from a project by the Syrian sculptor Assem Al-basha³. The project is to restore the damaged Al-Ma'arri⁴ Sculpture. The images are captured from a 2-minutes video⁵ by Aljazeera Arabic TV to report about the "najoon" project⁶.

4.1 Concrete example of test result

Fig. 12 shows the outcomes of two test steps for restoring image. 1 in Fig. 10. The outcomes are concrete example of the results we want to report in our oral conference presentation. Finally, in next weeks and before the ICERI conference we plan to conduct and test more experiments. The test results of these experiments will be shared during our oral presentation that aims to share reflections and to exchange feedback with the ICERI community.



Fig 12. Outcomes of two test steps for restoring image. 1 in Fig. 10. They are produced by the pencil tool available via the online link: <https://pincel.app/tools/inpaint>.

³ Assem Al-basha is one of the sons of the Syrian revolution and its participants. He believed in its justice and in a free Syria, and was active in achieving its goals. He is a Syrian survivor who went beyond the victim stage after leaving Syria, and turned to action as a Syrian in the diaspora and countries of asylum. As sculptor he completed Ras al-Ma'arri to be a symbol and a beacon for survivors. He hosted the Najoon team with his usual generosity throughout the days of working on the Najoon project. He is one of the founders of Najoon project and a member of its work team. <https://najoon.org/maarri/assembasha/>

⁴ Al-Ma'arri (December 973 – May 1057), also known by his Latin name Abulola Moarrensensis;^[2] was an Arab philosopher, poet, and writer from Ma'arrat al-Nu'man, Syria. Because of his controversially *irreligious worldview*, he is known as one of the "foremost atheists" of his time according to Nasser Rabbat: Tharoor, Kanishk; Maruf, Maryam (8 March 2016). "Museum of Lost Objects: The Unacceptable Poet". BBC News. <https://www.bbc.com/news/magazine-35745962>.amp, Last Retrieved 26.09.24.

⁵ The video is available via <https://www.youtube.com/watch?v=LCKr6aPVv9g>, last retrieved on 26.09.2024.

⁶ "najoon" Project. The project is named by Syrian word "najoon" which means "the survivors of the Syrian detention camps". With the will of the survivors of the Syrian detention camps and their passion for life and action, we founded the Al-Ma'arri Project, which we generated from our cognitive and historical roots, and from our belief that "there is no Imam but the mind," as Al-Ma'arri himself said. As our permanent call states, the project was our insistence on clinging to life and refusing to submit and acquiesce to the reality that they wanted for us. It was a restoration of the maternal bond between us and our motherland, "Syria," a free homeland based on pluralism and democracy. The project website is <https://najoon.org/maarri/>

5 CONCLUSIONS

We have presented our expressive-art based FA practice as inquiry in intercultural settings for liberation STEM education. The settings focus on target groups of repressed adult-migrants/students, migrated to Germany, due to the enduring oppressive structures and the current wars in Ukraine and Syria. The paper shows how the practice is conceptualized to enable two DM prototypes 1) A prototype for Ukrainian target group and 2) A prototype for Syrian target group. The two suggested DM prototypes are conceptualized to serve the repressed students and help them experiment DM inpainting algorithms to face and overcome the repressive barriers in their cultural context. The DM prototypes are grounded based on the theoretical concepts "Generative words", "Generative themes", "Critical consciousness" and "Reading the world precedes reading the word" suggested by the critical educator Paul Freire [1,2,3, 10]. To test the two DM prototypes, a test case is suggested as next steps of this work-in-progress paper.

ACKNOWLEDGEMENTS

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